

NeuroTrax SensorDB CAVIA

Project overview: cognitive assessments + large-scale smartphone SQL digital phenotyping

A growing pipeline for a rich digital biomarker model: baseline phenotype and T1-to-T2 change phenotype.

Data Scale First

Raw data scale

10 TB

approximate project storage scale

Approx. per patient

122 GB

10 TB divided across NeuroTrax patients

Assessment span

359 days

Patient and NeuroTrax Overview

Patients total

83

Patients with T1

83

Patients with T1 + T2

62

Patients with global delta

62

Cognitive patients with device label

82

Median T1-to-T2 gap

246 d

Phase 2 Work Underway

Target analysis tables

21

Reviewed table files

10

Selected features

29

Inventory rows

44

Two Main Project Outcomes

1

T1 baseline digital phenotype

Describe smartphone-derived behavior/context around the first cognitive assessment and later test prediction of NeuroTrax T1 score.

2

T1-to-T2 digital phenotype delta

Quantify digital changes between visits and later test whether they help predict NeuroTrax T2 cognitive outcome.

Experiment Time Span

First T1

2025-01-05

Last T2

2025-12-30

All-candidate span

359 d

NeuroTrax Feature Domains

Overall / Global

global_T1, global_T2, global_delta

Memory

memory_T1, memory_T2, memory_delta

Executive function

ef_T1, ef_T2, ef_delta

Attention

attention_T1, attention_T2, attention_delta

Processing speed

processing_speed_T1/T2/delta

Verbal

verbal_T1, verbal_T2, verbal_delta

Motor

motor_T1, motor_T2, motor_delta

IQ

iq_T1, iq_T2, iq_delta